

# Exercises from Ch. 10: Noetherian Rings and Modules, Lang

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Last Updated: July 2, 2026

**Disclaimer:** The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

## Exercises

**Exercise 0.1.** (Exercise 2)

Let  $\mathfrak{p}$  be a prime ideal, and  $\mathfrak{a}, \mathfrak{b}$  ideals of  $A$ . If  $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ , show that  $\mathfrak{a} \subseteq \mathfrak{p}$  or  $\mathfrak{b} \subseteq \mathfrak{p}$ .

*Proof.* Assume that  $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$  and  $\mathfrak{a} \not\subseteq \mathfrak{p}$ . Recall that

$$\mathfrak{a}\mathfrak{b} = \left\{ \sum a_i b_i : a_i \in \mathfrak{a}, b_i \in \mathfrak{b} \right\} \subseteq \mathfrak{a} \cap \mathfrak{b}.$$

Let  $x \in \mathfrak{a} \setminus \mathfrak{p}$  and let  $y \in \mathfrak{b}$ . Then  $xy \in \mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ . Since  $\mathfrak{p}$  is a prime ideal and  $x \notin \mathfrak{p}$ , it follows that  $y \in \mathfrak{p}$ . Since  $y \in \mathfrak{b}$  was arbitrary, we have  $\mathfrak{b} \subseteq \mathfrak{p}$ . □

For an ideal  $\mathfrak{a}$ , recall from Chapter IX, §5 that  $\mathcal{L}(\mathfrak{a})$  is the set of primes containing  $\mathfrak{a}$ .

**Exercise 0.2.** (Exercise 11)

Let  $A$  be a commutative ring and  $M$  an  $A$ -module. Define the **support** of  $M$  by

$$\text{supp}(M) = \{\mathfrak{p} \in \text{spec}(A) : M_{\mathfrak{p}} \neq 0\}.$$

If  $M$  is finite over  $A$ , show that  $\text{supp}(M) = \mathcal{L}(\text{ann}(M))$ , where  $\text{ann}(M)$  is the annihilator of  $M$  in  $A$ , that is the set of elements  $a \in A$  such that  $aM = 0$ .

*Proof.* Assume that  $M$  is finite over  $A$ , that is,  $M = \langle v_1, \dots, v_n \rangle$  with  $v_i \in M$  generators.   
( $\text{supp}(M) \subseteq \mathcal{L}(\text{ann}(M))$ ): Let  $\mathfrak{p} \in \text{supp}(M)$ . We need to show that  $\text{ann}(M) \subseteq \mathfrak{p}$ . Let  $a \in \text{ann}(M)$ , i.e.  $aM = \{ax : x \in M\} = 0$ . By assumption,

$$M_{\mathfrak{p}} = \{x/s : x \in M, s \notin \mathfrak{p}\} \neq 0.$$

Observe that  $aM_{\mathfrak{p}} = 0$ . By way of contradiction, assume that  $a \notin \mathfrak{p}$ . Then the image of  $a$  under the homomorphism  $A \xrightarrow{\varphi} A_{\mathfrak{p}}$  is invertible. Recall that localization preserves the property of being a module, that is,  $M_{\mathfrak{p}}$  is an  $A_{\mathfrak{p}}$ -module. Let  $u = \varphi(a) = a/1$ . Let  $m \in M_{\mathfrak{p}}$  be arbitrary. Then

$$m = (u^{-1}u)m = u^{-1}(um) = u^{-1} \cdot 0 = 0.$$

Hence,  $M_{\mathfrak{p}} = 0$ , contradicting  $\mathfrak{p} \in \text{supp}(M)$ . Therefore,  $a \in \mathfrak{p}$ , so  $\text{supp}(M) \subseteq \mathcal{L}(\text{ann}(M))$ .

( $\mathcal{L}(\text{ann}(M)) \subseteq \text{supp}(M)$ ): Let  $\mathfrak{p} \in \mathcal{L}(\text{ann}(M))$ . Hence,  $\mathfrak{p} \subseteq A$  is prime and  $\text{ann}(M) \subseteq \mathfrak{p}$ . We wish to show that  $M_{\mathfrak{p}} \neq 0$ . By way of contradiction, assume that  $M_{\mathfrak{p}} = 0$ . Thus, the image of every  $x \in M$  in  $M_{\mathfrak{p}}$  is 0. In particular, for all  $i = 1, \dots, n$  there exist  $s_i \notin \mathfrak{p}$  such that  $s_i v_i = 0$  (definition of being zero in the localization). Set

$$\bar{s} = \prod_{i=1}^n s_i.$$

Hence,  $\bar{s}$  annihilates every  $v_i$  and thus since  $M = \langle v_1, \dots, v_n \rangle$ ,  $\bar{s}$  annihilates  $M$ , that is,  $\bar{s} \in \text{ann}(M)$ . But since each  $s_i \notin \mathfrak{p}$  and  $\mathfrak{p}$  is prime,  $\bar{s} \notin \mathfrak{p}$  contradicting the assumption that  $\mathfrak{p} \in \mathcal{L}(\text{ann}(M))$ , i.e. the condition  $\text{ann}(M) \subseteq \mathfrak{p}$ . Therefore,  $M_{\mathfrak{p}} \neq 0$ , so  $\mathfrak{p} \in \text{supp}(M)$  and  $\mathcal{L}(\text{ann}(M)) \subseteq \text{supp}(M)$ . Thus,

$$\text{supp}(M) = \mathcal{L}(\text{ann}(M)).$$

□

**Exercise 0.3.** (Exercise 12)

Let  $A$  be a Noetherian ring and  $M$  an  $A$ -module. Let  $I$  be an ideal of  $A$  such that  $\text{supp}(M) \subseteq \mathcal{L}(I)$ . Then  $I^n M = 0$  for some  $n > 0$ .

*Proof.* By the previous exercise,

$$\text{supp}(M) = \mathcal{L}(\text{ann}(M)).$$

Thus, our assumption that  $\text{supp}(M) \subseteq \mathcal{L}(I)$  can be equivalently written as

$$\mathcal{L}(\text{ann}(M)) \subseteq \mathcal{L}(I).$$

By IX Proposition 5.1(iii),  $\mathcal{L}(\text{ann}(M)) \subseteq \mathcal{L}(I)$  if and only if  $\text{rad}(I) \subseteq \text{rad}(\text{ann}(M))$ . Observe that if  $x \in I$ , then  $x \in \text{rad}(I)$  trivially, and thus we have  $I \subseteq \text{rad}(I)$ , so, in particular,

$$I \subseteq \text{rad}(\text{ann}(M)).$$

Since  $A$  is Noetherian,  $I$  is finitely generated, say  $I = (x_1, \dots, x_m)$ . By the above containment, for each generator  $x_i$  there exists  $n(x_i) \geq 1$  such that  $x_i^{n(x_i)} \in \text{ann}(M)$ . Observe that  $I^k$  for some  $k \in \mathbb{Z}^+$  is generated by monomials

$$x_1^{w_1} \cdots x_m^{w_m}, \quad w_1 + \cdots + w_m = k.$$

We wish to find a  $k$  such that  $I^k \subseteq \text{ann}(M)$ . Define

$$k = \sum_{i=1}^m (n(x_i) - 1) + 1,$$

and we claim this choice of  $k$  induces the above containment. Every element of  $I^k$  is an  $A$ -linear combination of monomials  $x_1^{w_1} \cdots x_m^{w_m}$  of total degree  $k$ , i.e.  $\sum w_i = k$ . Hence, it suffices to show that each monomial is in  $\text{ann}(M)$ . Let  $x_1^{w_1} \cdots x_m^{w_m}$  be a monomial of total degree  $k$ . If  $w_i < n(x_i)$  for all  $i = 1, \dots, m$ , then  $w_i \leq n(x_i) - 1$  which implies  $w_1 + \cdots + w_m \leq \sum_{i=1}^m (n(x_i) - 1) = k - 1$  contradicting  $x_1^{w_1} \cdots x_m^{w_m}$  being a monomial of total degree  $k$ . Thus, there exists some  $i$  such that  $w_i \geq n(x_i)$ . Hence,  $x_1^{w_1} \cdots x_i^{w_i} \cdots x_m^{w_m} = x^{n(x_i)} (x_1^{w_1} \cdots x_i^{w_i - n(x_i)} \cdots x_m^{w_m})$ . Since  $x_i^{n(x_i)} \in \text{ann}(M)$ ,  $x_i^{n(x_i)} M = 0$ . Thus, for all  $m \in M$

$$(x_i^{n(x_i)} (x_1^{w_1} \cdots x_i^{w_i - n(x_i)} \cdots x_m^{w_m})) \cdot m = (x_1^{w_1} \cdots x_i^{w_i - n(x_i)} \cdots x_m^{w_m}) \underbrace{(x_i^{n(x_i)} m)}_{=0} = 0.$$

Thus, the monomial  $\prod x_i^{w_i}$  annihilates  $M$ , so  $\prod x_i^{w_i} \in \text{ann}(M)$ . Therefore, every generator of  $I^k$  lies in  $\text{ann}(M)$ , and thus  $I^k \subseteq \text{ann}(M)$ . Hence, we have found a  $k > 0$  such that  $I^k M = 0$ . □